Summary and Recommendation

Objective: The analysis primarily investigates key factors influencing customer churn across demographic attributes, contract terms, payment methods, and service add-ons.

Baseline Metrics: Out of 7,043 total customers analyzed, 1,869 customers churned, resulting in an overall churn rate of **26.54%** (73.46% retention rate).

Key Insights:

- **Contract Type:** Customers on Month-to-month contracts demonstrate a high churn rate of **42.71%** (1,655 out of 3,875 customers), accounting for **88.55% of all total churn**. In contrast, churn drops significantly to **11.27%** for One-Year contracts and **2.83%** for Two-Year contracts.
- **Payment Methods:** Customers paying via Electronic Check experience a **45.29% churn rate** (1,071 out of 2,365 customers). Automated payment methods show much higher retention, with Bank Transfer at **16.71% churn** (258 out of 1,544) and Credit Card at **15.24% churn** (232 out of 1,522).
- **Senior Citizens:** Senior citizens represent **16.21%** of the customer base but show a significantly elevated churn rate of **41.68%** (476 out of 1,142), compared to **23.61%** among non-seniors (1,393 out of 5,901).
- **Service Add-Ons & Internet Type:** Fiber Optic subscribers exhibit a **41.89% churn rate** (1,297 out of 3,096) compared to **18.96%** for DSL users. Having key protection services drastically improves retention:
    - **Tech Support:** Customers without Tech Support have a **41.64% churn rate**, compared to **15.17%** for those with it.
    - **Online Security:** Customers without Online Security have a **41.77% churn rate**, compared to **14.61%** for those with it.

Churn Rate by Tenure:

- Customers with short tenure (1 to 2 months) experience the highest rate of attrition (~50%+ churn rate).
- Churn rates drop below **20%** after 12 months and fall below **7%** for customers with over 60 months of tenure, emphasizing the critical importance of early-stage retention programs.

1. Demographic Breakdown

Senior Citizen Status

Senior citizens account for **16.21%** of the customer base but exhibit a disproportionately elevated churn risk compared to non-seniors.

Category	Total Count	% of Customer Base	Retained Count	Churned Count	Group Churn Rate	Group Retention Rate
Non-Senior Citizen	5,901	83.79%	4,508	1,393	**23.61%**	76.39%
Senior Citizen	1,142	16.21%	666	476	**41.68%**	58.32%

Gender

Gender distribution is nearly evenly split and demonstrates negligible variance in churn behavior.

Category	Total Count	% of Customer Base	Retained Count	Churned Count	Group Churn Rate	Group Retention Rate
Female	3,488	49.52%	2,549	939	**26.92%**	73.08%
Male	3,555	50.48%	2,625	930	**26.16%**	73.84%

2. Contract Commitment & Financial Profile

Contract Type Impact

Contract duration serves as the single strongest structural predictor of retention. **55.02%** of all customers are on short-term Month-to-Month contracts, which drive **88.55% of total churn volume** (1,655 out of 1,869 churned customers).

Contract Type	Total Count	% of Total Base	Retained Count	Churned Count	Group Churn Rate	Group Retention Rate
Month-to-Month	3,875	55.02%	2,220	1,655	**42.71%**	57.29%
One Year	1,473	20.91%	1,307	166	**11.27%**	88.73%
Two Year	1,695	24.07%	1,647	48	**2.83%**	97.17%

Financial Metrics Summary

- **Monthly Charges:** Mean of **$64.76** (Std: $30.09, Range: $18.25 to $118.75). Higher monthly billing amounts closely align with elevated churn risk, particularly when unbundled.
- **Total Charges:** Mean of **$2,279.73** (Std: $2,266.79). 11 accounts registered $0 total charges due to 0-month tenure; these were converted to numeric zeros during data cleaning.

3. Payment Method Analysis

Electronic Check is the dominant payment method among churned users, carrying a churn rate nearly **2.7x higher** than automated payment channels.

Payment Method	Total Count	% of Total Base	Retained Count	Churned Count	Group Churn Rate	Group Retention Rate
Electronic Check	2,365	33.58%	1,294	1,071	**45.29%**	54.71%
Mailed Check	1,612	22.89%	1,304	308	**19.11%**	80.89%
Bank Transfer (auto)	1,544	21.92%	1,286	258	**16.71%**	83.29%
Credit Card (auto)	1,522	21.61%	1,290	232	**15.24%**	84.76%

4. Internet Infrastructure & Service Add-On Impact

Internet Service Type

Fiber Optic subscribers represent **43.96%** of total customers but experience an elevated **41.89% churn rate**, driven primarily by high monthly pricing (70–110/mo) without adequate support features attached.

Internet Service	Total Count	% of Total Base	Retained Count	Churned Count	Group Churn Rate	Group Retention Rate
Fiber Optic	3,096	43.96%	1,799	1,297	**41.89%**	58.11%
DSL	2,421	34.37%	1,962	459	**18.96%**	81.04%
No Internet	1,526	21.67%	1,413	113	**7.40%**	92.60%

Key Support Add-Ons

Subscribing to protection features reduces churn rates by **over 60%** relative to users without these features enabled:

- **Tech Support:**
  - **No Tech Support: 41.64% churn rate** (1,446 churned out of 3,473)
  - **With Tech Support: 15.17% churn rate** (310 churned out of 2,044)
  - **No Internet Service: 7.40% churn rate** (113 churned out of 1,526)
- **Online Security:**
  - **No Online Security: 41.77% churn rate** (1,461 churned out of 3,498)
  - **With Online Security: 14.61% churn rate** (295 churned out of 2,019)
- **Online Backup:**
  - **No Online Backup: 39.93% churn rate** (1,233 churned out of 3,088)
  - **With Online Backup: 21.53% churn rate** (523 churned out of 2,429)

5. Tenure Lifecycle Dynamics

Average customer tenure is **32.37 months**. Early-tenure friction dominates total attrition:

- **0 to 2 Months Tenure:** Accounts for the highest churn velocity (~50%+ churn rate during the first 60 days).
- **12+ Months Tenure:** Churn rate drops below **20%**.
- **60+ Months Tenure:** Churn rate drops below **7%**, establishing a stable core customer base.

Core Retention Strategy Matrix

1. **Targeted First-90-Days Onboarding:** Focus retention programs on new Fiber Optic and Month-to-Month subscribers within their first 3 months to mitigate early tenure drop-offs.
2. **Contract Transition Campaign:** Offer a 5% to 10% billing discount or a free service add-on to transition Month-to-Month customers (42.71% churn) into 1-Year or 2-Year plans (11.27% and 2.83% churn).
3. **Automated Payment Migration:** Incentivize the 33.58% of customers paying via Electronic Check (45.29% churn) to adopt automated credit card or bank transfer payments (15.24%–16.71% churn).
4. **Security & Tech Support Bundling:** Default Fiber Optic plans with bundled Tech Support and Online Security to lower high standalone Fiber Optic attrition.

Recommendations

1. **Onboarding & Early Tenure Focus:** Implement a targeted 90-day onboarding program with special promotional offers or proactive customer care outreach to reduce 1-to-2 month churn.
2. **Contract Migration Incentives:** Provide incentives (e.g., discounted monthly rates or service upgrades) to transition month-to-month subscribers onto 1-Year or 2-Year plans.
3. **Automated Payment Drive:** Encourage migration away from Electronic Checks toward automatic bank transfers or credit card payments by offering minor billing discounts.
4. **Service Bundling:** Package **Online Security** and **Tech Support** into core internet plans, particularly for new Fiber Optic subscribers and Senior Citizens, to increase product stickiness.